

On a Derivable Formula for Long-Wave Radiation From Clear Skies

WILFRIED BRUTSAERT

School of Civil and Environmental Engineering, Cornell University, Ithaca, New York 14853

A derivation is presented for the effective atmospheric emissivity to predict downcoming long-wave radiation at ground level under a clear sky and for a nearly standard atmosphere. The results are in good agreement with those obtainable with empirical formulae based on water vapor pressure and temperature. However, the proposed formulation has the advantage that its simple functional form is based on physical grounds without the need for empirical parameters from radiation measurements. Also, in contrast to the empirical equations, it may be adjusted in a simple way to reflect changes in climatic and atmospheric conditions.

INTRODUCTION

The accurate determination of the low temperature or terrestrial radiation at any point in the atmosphere is a problem of great practical and theoretical complexity. Therefore considerable efforts have been made to develop simplified methods to calculate atmospheric radiation. These calculations can often be conveniently carried out by means of radiation charts [e.g., *Haltiner and Martin*, 1957; *Goody*, 1964; *Kondratyev*, 1969] that have appeared in the literature. The simplification consists essentially of assuming an isotropic source function in a stratified atmosphere for which absorption coefficient and temperature are functions of elevation only. This then allows integration of Schwarzschild's transfer equation over all directions and over all wavelengths, through the definition of the slab emissivity, to yield for the net radiative flux at reference z [e.g., *Goody*, 1964]

$$F_L(z) = F_{LU}(z) - F_{LD}(z) \quad (1)$$

The terms on the right are the upward and downward flux, respectively, each of which is given by

$$(F_{LU}, F_{LD}) = \int_{a=0}^{\infty} \pi B(T) \frac{\partial \epsilon_c(1.66a, T)}{\partial a} da \quad (2)$$

where $B(T) = \sigma T^4/\pi$ is the integrated Planck function, $T = T(z)$ the temperature, σ the Stefan-Boltzmann constant, and ϵ_c the column emissivity relating the emission of a column of gas to that of a blackbody. The quantity a is the scaled amount of matter or in the case of cloudless sky mainly the water vapor in the air column measured from the level z down in F_{LU} and up in F_{LD} and scaled for the pressure effect. This pressure correction is often taken as a square root as was suggested by Elsasser [e.g., *Haltiner and Martin*, 1957], so that the scaled amount of water vapor is in differential form:

$$da = \rho_w(p/p_s)^{1/2} dz \quad (3)$$

where p and p_s are the pressures at levels $z = z$ and $z = z_s$, respectively.

Equations (1), (2), and (3); or very similar expressions, constitute the basis of the numerous radiation chart procedures that are available. These charts are quite satisfactory for most practical problems.

Unfortunately, however, it is not always easy to obtain the necessary atmospheric temperature and humidity profiles, $T(z)$ and $\rho_w(z)$, that are required for the computation of the radiative flux. Thus for certain practical design purposes or for

retrospective estimates of radiation, use is often made of empirical formulae that provide downward atmospheric radiation at ground level in terms of meteorological data near the surface. This is usually done by an equation of the type

$$R_{LD} = \epsilon_a \sigma T_0^4 \quad (4)$$

where T_0 is the air temperature near the ground (at shelter level) and ϵ_a may be called the effective atmospheric emissivity, which is taken to be a function of T_0 or of the vapor pressure e_0 or of both.

In what follows, a brief analysis is presented of the downward flux for a simplified atmosphere in order to throw some light on the theoretical validity of obtaining atmospheric radiation by means of (4) in terms of near-surface temperature and humidity data.

COMPUTATION FOR STANDARD CONDITIONS

It is possible to obtain an analytical solution of (1) and (2) by assuming a nearly standard atmosphere and an algebraic relationship for the emissivity. In most model atmospheres the temperature decreases with a constant lapse rate γ up to a level H above which it remains approximately constant. For example, in the case of the ICAO standard atmosphere [*U.S. Committee on Extension to the Standard Atmosphere*, 1962], $\gamma = -6.5^\circ\text{K/km}$, and $H = 11$ km; at sea level the temperature is $T_0 = 288.15^\circ\text{K}$, and for $11 \text{ km} \leq z \leq 20 \text{ km}$ it is constant at 216.65°K . Presumably, for the present analysis, mainly the lowest 5–6 km are of concern. Therefore to simplify the determination of the integrals at the upper limit for $z = \infty$, the temperature may be assumed to be given by

$$T = T_0 \exp [-(\gamma/T_0)z] \quad (5)$$

which with (γ/T_0) being of the order of $2.26 \times 10^{-2} \text{ km}^{-1}$ should be adequate for the lowest 15 km.

The vertical pressure profile in a dry atmosphere with a linear temperature profile is by virtue of the hydrostatic law and the equation of state

$$p = p_0(T/T_0)^{g/\gamma R_d} \quad (6)$$

Thus with (5) this produces approximately

$$p = p_0 \exp [-(g/R_d T_0)z] \quad (7)$$

where p_0 is the pressure at sea level, g the acceleration of gravity, and R_d the gas constant for dry air. With $(g/R_d T_0)$ equal to approximately 0.13 km^{-1} , (7) should be adequate for the present purpose up to levels of 5–10 km.

The distribution of vapor pressure e in the lowest 10 km is

also often described by an exponential relationship. For example, Yamamoto [1949] proposed

$$e = e_0 \exp \{-[5.8 \times 10^3(\gamma/T_0^2) + 0.055]z\}$$

where z is in kilometers. Thus on the basis of (5) and the equation of state the water vapor density is similarly an exponential function given by

$$\rho_w = (0.622e_0/R_dT_0) \exp(-k_w z) \quad (8)$$

where k_w is a constant depending on the atmospheric conditions. In the case of Yamamoto's hypothesis, one has $k_w = [5.8 \times 10^3(\gamma/T_0^2) - (\gamma/T_0) + 0.055]$. When $T_0 = 290^\circ\text{K}$ and $\gamma = -6^\circ\text{K/km}$, this would yield $k_w = 0.45 \text{ km}^{-1}$; this is quite close to $k_w = 0.44 \text{ km}^{-1}$, which Reitan [1963] found to be a representative value for the United States.

Emissivity data based on actual atmospheric measurements and on calculations have been published by many authors [e.g., Yamamoto, 1952; Kondratyev, 1969]. Plotting these data on log-log paper, one finds that they can conveniently be approximated by a power function

$$\epsilon_s = Aa^m \quad (9)$$

where $\epsilon_s(a) = \epsilon_c(1.66a)$ is the emissivity of a slab of water vapor with CO_2 . The value of the parameters depends somewhat on the range of a . In the range $0.01 < a < 1.0 \text{ cm}$ of water the values appear to lie around $A = 0.75$ and $m = 1/6$, whereas for $0.1 < a < 10 \text{ cm}$, $A = 0.75$ and $m = 1/8$ should be satisfactory.

With these functions for the temperature, pressure, and humidity profiles and for the emissivity it is possible to integrate (2). Thus substitution of (5), (7), (8), and (9) gives

$$R_{LD} = F_{LD}(0) = \sigma T_0^4 \left(\frac{0.622e_0}{R_d T_0} \right)^m \cdot \int_0^\infty e^{-k_1 z} \left(\int_0^z e^{-k_2 y} dy \right)^{m-1} dz$$

where $k_1 = [(4\gamma/T_0) + k_3]$ and $k_2 = (k_w + 0.065)$. Integration results in

$$R_{LD} = \sigma T_0^4 m A \left(\frac{0.622e_0}{k_2 R_d T_0} \right)^m B\left(\frac{k_1}{k_2}, m\right) \quad (10)$$

where $B(\quad)$ is the beta function [e.g., Abramowitz and Stegun, 1964]. Substitution of the above-mentioned typical values of $k_w = 0.44$, $m = 1/7$, $A = 0.75$, and $(\gamma/T_0) = 0.0226 \text{ km}^{-1}$ (for k_1) yields immediately (if e_0 is in millibars and T_0 is in degrees Kelvin)

$$R_{LD}/(\sigma T_0^4) = \epsilon_{a0} = 1.24(e_0/T_0)^{1/7} \quad (11)$$

where ϵ_{a0} is the effective emissivity for clear sky; note that the value of the coefficient 1.24 is relatively insensitive to variations in the 'typical' values of the parameters. Because the right-hand side of (11) is very insensitive to changes in T_0 , it

can also be written for a typical shelter temperature $T_0 = 288^\circ\text{K}$ as

$$\epsilon_{a0} = 0.553e_0^{1/7} \quad (11')$$

where again e_0 is in millibars.

Although the assumed vertical profiles represent actual conditions in the atmosphere probably for brief periods only, they are close approximations of averages on a weekly or monthly basis. Thus (11) and (11') may be expected to yield satisfactory results for representative weather conditions under clear skies. Because they only require the temperature and the vapor pressure, they should be useful in obtaining a reasonable estimate where extreme accuracy is not required. To confirm their validity, they will next be compared with several empirical formulae.

COMPARISON WITH EMPIRICAL FORMULAE

One of the better-known formulae used in practice is due to Brunt [1932], namely, (4) with

$$\epsilon_{a0} = a_B + b_B e_0^{1/2} \quad (12)$$

where a_B and b_B are coefficients to be determined from observational data. Brunt based the form of his formula on the fact that the solution of unsteady heat conduction problems often involves the square root of the thermal diffusivity. Since he presumed that radiative transfer in the atmosphere is analogous to molecular conduction, the coefficient replacing the diffusivity was shown to be inversely proportional to the vapor pressure. Therefore he took a linear, that is, the simplest possible relationship between atmospheric radiation and the square root of the vapor pressure. Although the diffusion analogy has severe limitations [e.g., Kondratyev, 1969], this linear relationship has been widely used with good success. Perhaps more than on any other formula much work has been done by numerous investigators to determine its parameters. Some of these results are shown in Table 1. Yamamoto [1950] and Kondo [1967] found that $a_B = 0.51$ and $b_B = 0.066$, obtained from a mean curve through data by several other authors, would be satisfactory.

It can be seen in Figure 1 that the simplified theoretical result of (11) is in excellent agreement with the various estimates of Brunt's formula. But (11) has the advantage that only one parameter is required, and in the absence of experimental data it may be estimated from profile parameters in (10).

A second formula that merits comparison is that obtained by Swinbank [1963]:

$$\epsilon_{a0} = 0.398 \times 10^{-5} T_0^{2.148} \quad (13)$$

which he rounded off to

$$\epsilon_{a0} = 0.92 \times 10^{-5} T_0^2 \quad (13')$$

Albeit the basis of this formulation appears to be purely em-

TABLE 1. Some Values of the Constants in Brunt's Formula (With e_0 in Millibars)

Reference	Location	Latitude	Altitude, m	a_B	b_B	Correlation	Period
Brunt [1932]	Benson, England	52°N	6	0.52	0.065	0.97	monthly
Goss and Brooks [1956]	Davis, Central Valley, California	38°N	14	0.66	0.039	0.89	monthly
Anderson [1954]	Lake Hefner, Oklahoma	36°N	369	0.68	0.036	0.92	monthly
DeCoster and Schuepp [1957]	Kinshasa, Zaire	4.8°N	321	0.645	0.048	...	daily

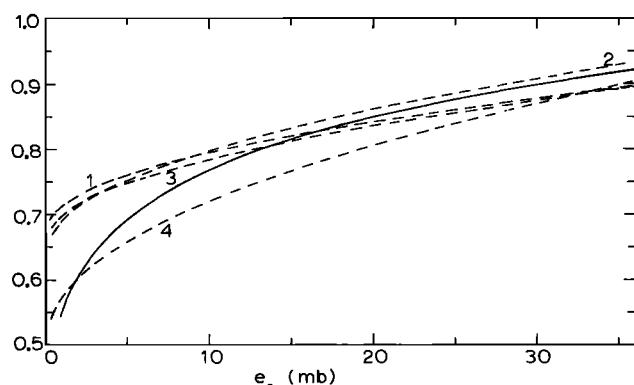


Fig. 1. The dependence of the effective atmospheric emissivity ϵ_{a0} on the vapor pressure near the surface at screen level e_0 (in millibars) as calculated by means of (11) for $T_0 = 288^\circ\text{K}$ (solid line) and by means of Brunt's formula (dashed lines). Curve 1 is based on the parameters obtained by Anderson [1954], curve 2 on those of DeCoster and Schuepp [1957], curve 3 on those of Goss and Brooks [1956], and curve 4 on those of Yamamoto [1950] and Kondo [1967].

irical, it can also be reconciled with the theoretical result of (11). Indeed, it has been found [Deacon, 1970] from monthly means at locations covering a considerable climatic range that the amount of precipitable water is proportional to the 16.8 power of the air temperature at screen level. Thus according to (8) one has $e_0 \sim T_0^{17.8}$, and substitution in (11) shows that $\epsilon_{a0} \sim T_0^{2.4}$, which is not very different from $T_0^{2.148}$.

SUMMARY AND CONCLUSIONS

A simple equation (equation (10)) has been derived analytically to compute downward long-wave radiation at ground level under clear skies and nearly standard atmospheric conditions. It compares favorably with several other well-known empirical formulae such as the formula of Brunt and that of Swinbank. This agreement indicates that albeit the calculated result is probably not very sensitive to the type of the adopted model, the assumed atmospheric functions (equations (5), (7), (8), and (9)) are quite suitable for the present purpose. Conversely, this may also explain why most empirical formulae proposed in the literature have been found satisfactory for long-term mean calculations.

The functional form of the available empirical equations was usually arrived at by mere inspection of the data and by correlation techniques. In contrast, the functional form of (10) and (11) is based on physical grounds, since it is obtained by

direct solution of the governing Schwarzschild equation. Thus under standard conditions, (10) requires no empirical parameters for its application, whereas the empirical formulae require at least one [Swinbank, 1963] or more [e.g., Brunt, 1932; Anderson, 1954] parameters to be determined from radiation experiments. Still, the main advantage of the present result (equation (10)) over the empirical equations is that it may be useful to determine rapidly local adjustments in the effective atmospheric emissivity ϵ_{a0} resulting from changes in climatic and simple atmospheric parameters.

REFERENCES

- Abramowitz, M., and I. A. Stegun (Eds.), *Handbook of Mathematical Functions*, in *Appl. Math. Ser.*, vol. 55, National Bureau of Standards, Washington, D. C., 1964.
- Anderson, E. R., Energy budget studies, Water Loss Investigations: Lake Hefner Studies, Technical Report, *U.S. Geol. Surv. Prof. Pap.* 269, 71-119, 1954.
- Brunt, D., Notes on radiation in the atmosphere, 1, *Quart. J. Roy. Meteorol. Soc.*, 58, 389-420, 1932.
- Deacon, E. L., The derivation of Swinbank's long-wave radiation formula, *Quart. J. Roy. Meteorol. Soc.*, 96, 313-319, 1970.
- DeCoster, M., and W. Schuepp, Mesures de rayonnement effectif à Leopoldville, *Acad. Roy. Sci. Colon. Brussels Bull. Seances*, 3, 642-651, 1957.
- Goody, R. M., *Atmospheric Radiation*, 436 pp., Clarendon, Oxford, 1964.
- Goss, J. R., and F. A. Brooks, Constants for empirical expressions for downcoming atmospheric radiation under cloudless sky, *J. Meteorol.*, 13, 482-488, 1956.
- Haltiner, G. J., and F. L. Martin, *Dynamical and Physical Meteorology*, McGraw-Hill, New York, 1957.
- Kondo, J., Analysis of solar radiation and downward long-wave radiation data in Japan, *Sci. Rep. Tohoku Univ., Ser. 5*, 18, 91-124, 1967.
- Kondratyev, K. Ya., *Radiation in the Atmosphere*, 912 pp., Academic, New York, 1969.
- Reitan, C., Surface dew point and water vapor aloft, *J. Appl. Meteorol.*, 2, 776-779, 1963.
- Swinbank, W. C., Long-wave radiation from clear skies, *Quart. J. Roy. Meteorol. Soc.*, 89, 339-348, 1963.
- U.S. Committee on Extension to the Standard Atmosphere, U.S. standard atmosphere, 1962, U.S. Govt. Print. Office, Washington, D. C., 1962.
- Yamamoto, G., Average vertical distribution of water vapour in the atmosphere, *Sci. Rep. Tohoku Univ., Ser. 5*, 1, 76-79, 1949.
- Yamamoto, G., On nocturnal radiation, *Sci. Rep. Tohoku Univ., Ser. 5*, 2, 27-43, 1950.
- Yamamoto, G., On a radiation chart, *Sci. Rep. Tohoku Univ., Ser. 5*, 4, 9-23, 1952.

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